# Sphere and ellipse fitting for camera calibration

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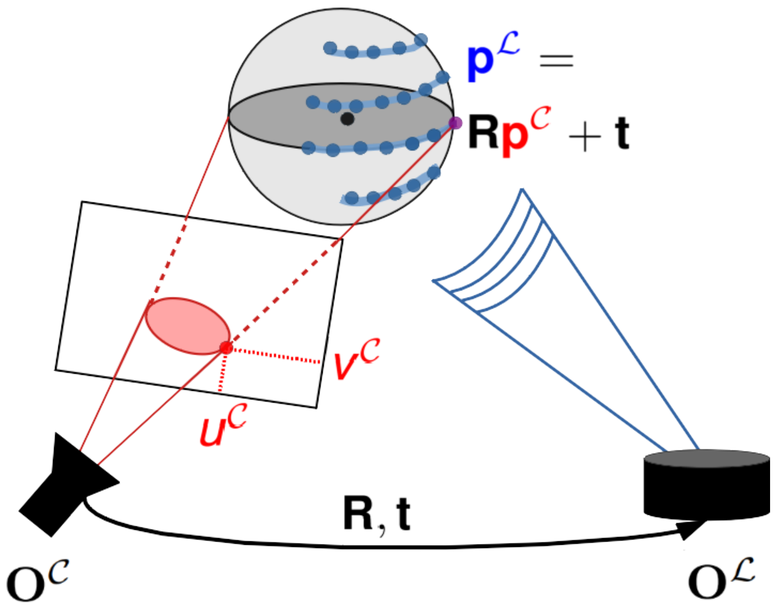
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## Motivation

The mounted sensor setup of the university test car: five Hikvison digital cameras (three normal lenses forward; two fisheye lenses sideways) and a Velodyne VLP-16 LiDAR.




## Basic Concept




$$\begin{bmatrix} \mathbf{p}^{\mathcal{L}} \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{R}_{3\times3} & \mathbf{t}_3 \\ \mathbf{0}_3^T & 1 \end{bmatrix}}_{\mathbf{T}^{\mathcal{LC}}} \begin{bmatrix} \mathbf{p}^{\mathcal{C}} \\ 1 \end{bmatrix}$$

In our former paper [1], we presented how the spheres are suitable for LiDAR-camera calibration which has three main part: (i) pointcloud based sphere center estimation, (ii) image-based sphere center estimation, and (iii) relative pose estimation by pointset registration. The aim of this poster is to present the theoretical improvements in the image-based sphere recognition.

## References

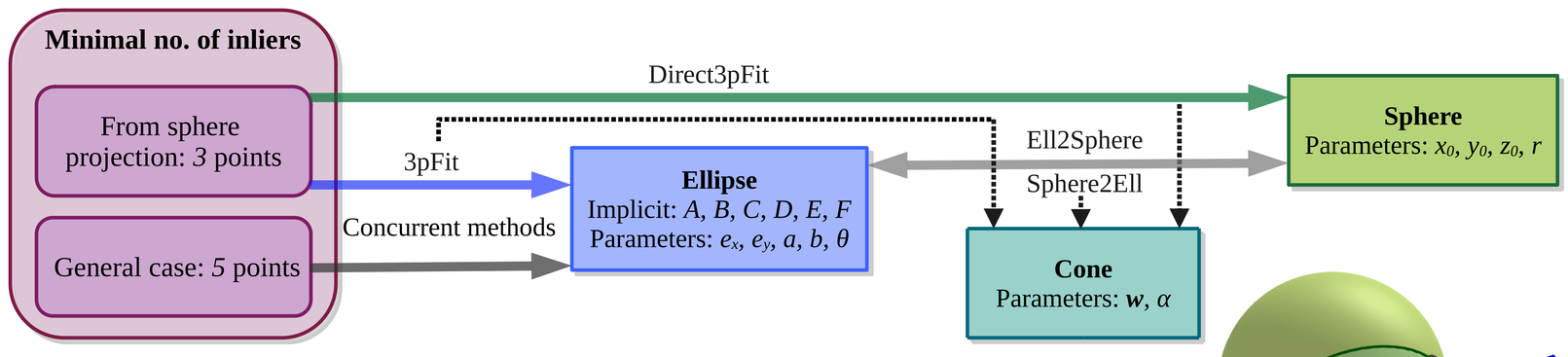
[1] Toth, T., Pusztai, Z., Hajder, L., *Automatic LiDAR-Camera Calibration of Extrinsic Parameters Using a Spherical Target*, 2020 IEEE International Conference on Robotics and Automation (ICRA), (2020), 8580-8586.

## Acknowledgements

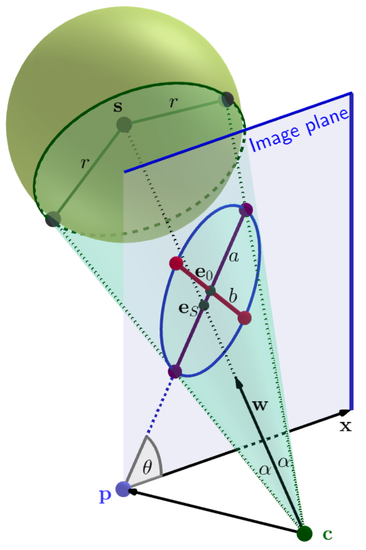
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## Proposed algorithms




This poster shows how to accurately estimate the projected ellipse parameters from contour points for calibrated cameras. Furthermore, we reveal how to evaluate the sphere parameters analytically. The proposed approaches require a minimum of three points exploiting the new constraints that we introduce. It is proven that three contour points of an ellipse are sufficient to estimate the projective line of the sphere centers. Hence, if the radius of the sphere is known, the sphere position can be estimated. As a result, novel geometry-based *three-point algorithms* are proposed for both minimal and overdetermined cases. Finally, the sphere parameters can be computed if the contour ellipse is known.




## Three-point solutions & the two constraints

*Axial constraint*: The line of the major ellipse axis intersects the principal point $\mathbf{p}$.

*Projectional constraint*: The connection between the ellipse center and the semi-axes:

$$\frac{a^2}{b^2} = \frac{e_x^2 + e_y^2}{b^2 + 1} + 1.$$

*Cone parameters*: Given at least three $\mathbf{p}_i$ points with $\mathbf{q}_i$ directions from $\mathbf{c}$: $\langle \mathbf{q}_1, \mathbf{w} \rangle = \langle \mathbf{q}_2, \mathbf{w} \rangle = \langle \mathbf{q}_3, \mathbf{w} \rangle = \ldots = \cos\alpha$. In matrix form: $\frac{1}{\cos\alpha}\mathbf{w} = \mathbf{M}^{+T}\mathbf{1}_3$, where $\mathbf{M} = [\mathbf{q}_1\ \mathbf{q}_2\ \mathbf{q}_3\ \ldots]$. The magnitude and the direction of $\mathbf{M}^{+T}\mathbf{1}_3$ gives $\alpha$ and $\mathbf{w}$.

*Sphere center*: If the cone parameters are defined: $S = \frac{r}{\sqrt{1-\cos^2\alpha}}\mathbf{w}$.

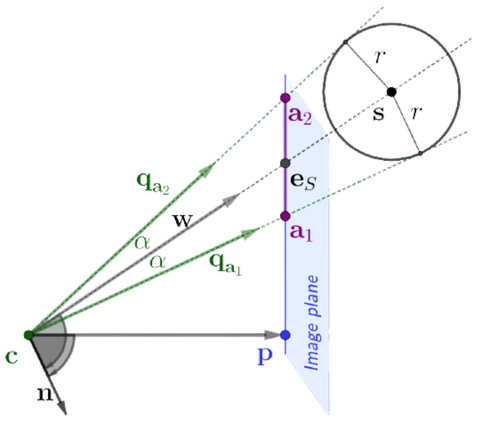
$$\mathbf{n} = \begin{cases} \begin{bmatrix} -w_y & w_x & 0 \end{bmatrix}^T \cdot \frac{1}{\sqrt{w_x^2 + w_y^2}} & \text{for } \mathbf{w} \neq \mathbf{p} \\ \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T & \text{for } \mathbf{w} = \mathbf{p} \end{cases}$$

$$\Downarrow$$

$$\mathbf{q}_{\mathbf{a}_1} = \mathbf{R}(\mathbf{n}, \alpha) \cdot \mathbf{w} \qquad \mathbf{q}_{\mathbf{a}_2} = \mathbf{R}(\mathbf{n}, \alpha)^T \cdot \mathbf{w}$$

$$\Downarrow \qquad \Downarrow$$

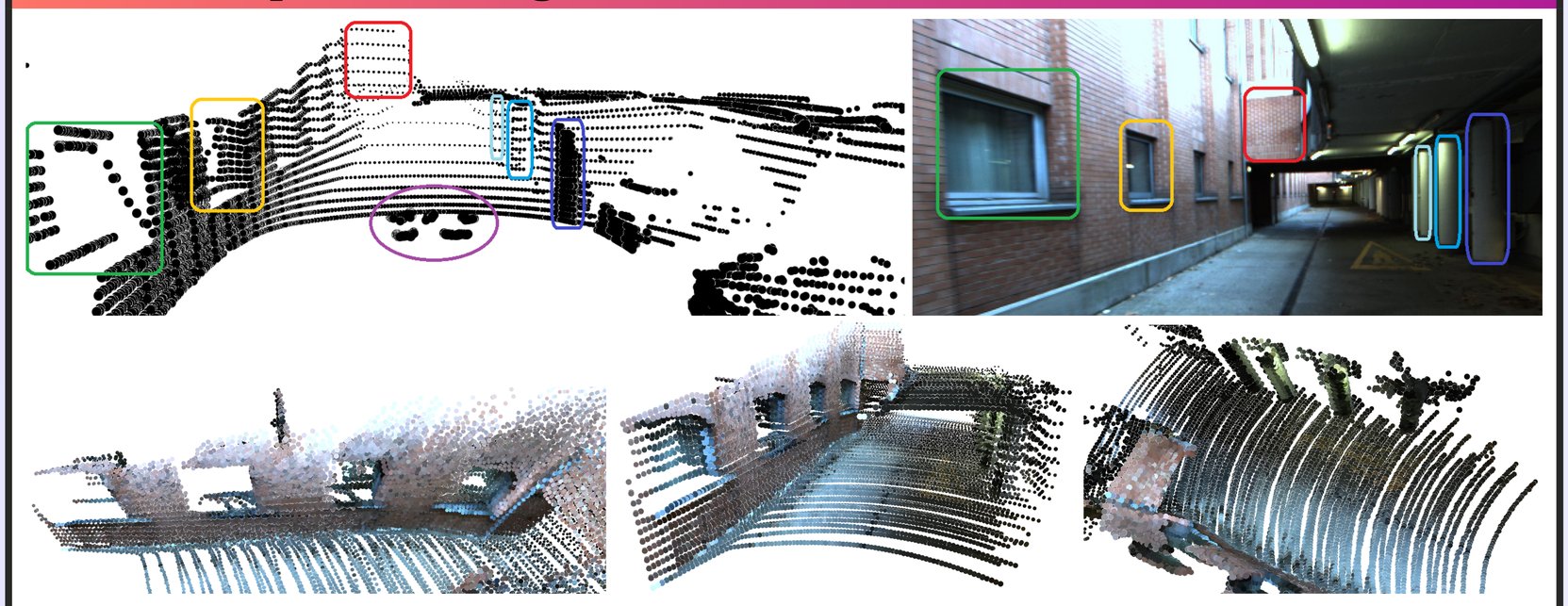
$$\mathbf{a}_1 = \Pi_{\mathcal{I}}(\mathbf{q}_{\mathbf{a}_1}) \qquad \mathbf{a}_2 = \Pi_{\mathcal{I}}(\mathbf{q}_{\mathbf{a}_2})$$




*Ellipse parameters*: If the major ellipse axis end-points are calculated, and $\Omega = e_x^2 + e_y^2 + 1 - a^2$ is introduced, then the ellipse parameters:

$$\mathbf{e}_0 = \frac{\mathbf{a}_1 + \mathbf{a}_2}{2}, \quad a = \frac{\|\mathbf{a}_1 - \mathbf{a}_2\|}{2}, \quad \theta = \operatorname{atan2}(e_y, e_x), \quad b = \sqrt{\frac{-\Omega + \sqrt{\Omega^2 + 4a^2}}{2}}.$$

## Colorized pointset registration



To validate the sphere-based calibration results, we registered a colored pointcloud sequence with the iterative closest point algorithm (bottom; same point cloud from different points of view) and displayed one original snapshot of the scene (top left: pointcloud; top right: camera image). On the top, the framed regions with the same color cover similar areas and are easy to identify in the colored 3D points. The points circled with purple belong to the ego car. On the bottom, the coloring around the window sills and the eaves (bottom left) exemplify the precision of the extrinsic calibration parameters, including the image-based sphere estimation. The dividing black line (bottom middle) and yellow road markings (bottom right) are also recognizable, and the road is in shadow after the third pillar (bottom right), like in the camera image. The main pillars with some intense greenish light are also observable.